

Technology Stack

Date	July7, 2026
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Project Name	Caption AI: AI-Powered Social Media Caption Generator

Technology Stack:

The following technologies were used to design, develop, test, and deploy the CaptionAI application.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Bootstrap 5	Used to create a responsive and user-friendly interface where users can enter campaign details, select a platform and tone, and view the generated captions, hashtags, and CTA suggestions.
2	Backend / Server-Side	Python 3.10+, Flask Framework	Flask was used to build the backend application. It processes user requests, communicates with the Groq API, and returns the generated content to the frontend.
3	Data / Configuration	.env, python-dotenv	Used to securely store the Groq API key and manage environment variables without exposing sensitive information in the source code.
4	Deployment	Ngrok	Used to create a secure public URL for the locally running Flask application, making it accessible for demonstration and testing.
5	Version Control	Git, GitHub	Used to manage source code, track project changes, and maintain the project repository throughout development.

6	AI Service / Third-Party API	Groq API (LLaMA 3.3-70B Versatile Model)	Used to generate AI-powered captions, hashtags, and CTA suggestions based on the user's campaign description, selected platform, and writing tone.
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